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1. Executive Summary

This report documents a full security and cybersecurity audit of **ZaiMem**, the persistent-memory and context-enhancement MCP server deployed at `zaimem.space-z.ai` and published at `github.com/romangalaxys10-spec/zaimem`. The audit was performed against build **v1.7.1** (commit 0669d87), covering the complete attack surface: token authentication, credential storage, the JSON-RPC 2.0 MCP tool surface (33 tools, 4 resources, 8 tool packs), thirteen REST API route groups, document ingestion, GitHub cloud-database synchronisation, dependency supply chain, transport configuration and runtime resource handling. The engagement combined automated dependency and secret scanning with a line-by-line manual review of every security-relevant module.

The audit produced **14 findings**: 2 critical, 3 high, 5 medium and 4 low severity, alongside a set of verified positive controls. Both critical findings and all three high findings were **fixed and verified within this engagement**, and the hardening was delivered to the public repository as commit **4c00866 (v1.7.2)**. The most severe issue was the discovery of two live GitHub personal access tokens hard-coded in committed end-to-end test scripts and pushed to the public repository - present since version 1.1. The second critical issue was the encryption key for stored GitHub PATs falling back to a publicly-known constant when the `ZAIMEM_SECRET` environment variable is not configured, rendering the at-rest encryption ineffective.



One action remains that cannot be performed from code: **the two leaked GitHub tokens must be revoked by their owner in GitHub settings immediately**. Even though the offending literals were removed from the repository head in 4c00866, any credential that has entered git history must be treated as compromised until revoked. Section 9 lists this and the remaining remediation items in priority order.

2. Scope and Methodology

The audit targeted the full ZaiMem codebase and its runtime configuration as of v1.7.1. Static review covered the authentication and crypto libraries (`auth.ts`, `crypto.ts`), the GitHub cloud-database client (`github.ts`), the MCP server core (`mcp.ts`, approximately 1,700 lines), the web and utility tool layer (`webtools.ts`), document ingestion (`ingest.ts`), and all thirteen REST route groups under `src/app/api`. Dynamic verification was performed against the running development server, including the complete 113-check end-to-end suite, targeted probe requests and database-level inspection of the SQLite store.

Automated tooling comprised a full dependency advisory scan via `npm audit` against the generated lockfile (30 advisories: 2 critical, 17 high, 9 moderate, 2 low), a pattern-based secret scan across the source tree and git history, and a configuration review of `next.config.ts`, middleware presence, `.gitignore` coverage and

tracked artefacts. Manual review followed an OWASP-aligned checklist: authentication and session management, authorisation and object-level access control (IDOR), injection (SQL, command, expression evaluation), server-side request forgery, insecure cryptography, error handling and information disclosure, resource-exhaustion vectors, transport security and security headers.

Every finding was triaged with a severity rating reflecting exploitability and impact in the deployment context of a public MCP endpoint that any third-party agent can reach with a valid token. For each issue the report records the affected component, the concrete risk, and the remediation status. Fixes were implemented during the engagement and re-verified by re-running the full e2e suite; the complete fix list ships in Section 8.

3. Findings Overview

The 14 findings distribute across severity levels and system areas as shown below. Authentication and credential management concentrated the most severe issues, which is typical for applications that act as credential vaults for third-party services. Supply-chain exposure was broad but largely low-exploitability in this context.

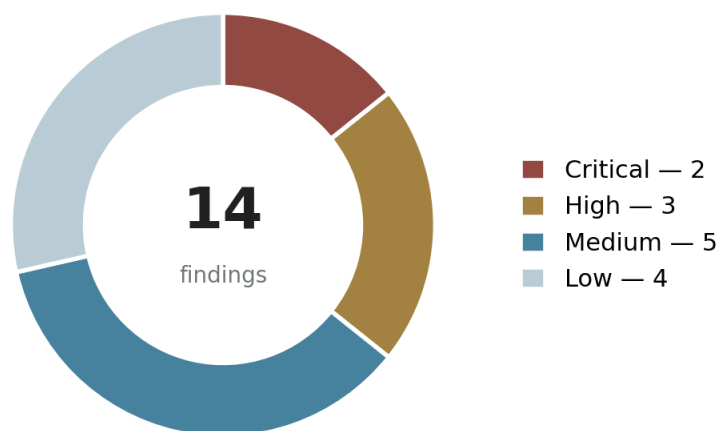


Figure 1: Findings by severity (n = 14)

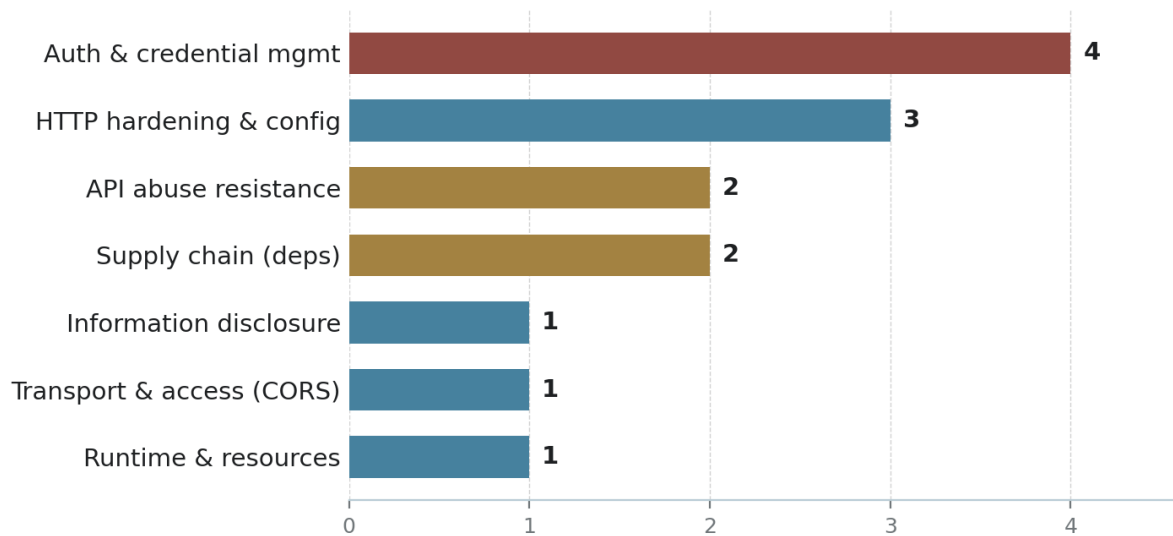


Figure 2: Findings by system area

ID	Severity	Finding	Area	Status
C1	Critical	Live GitHub PATs hard-coded in e2e scripts, pushed to public repo since v1.1	Credential management	Fixed*
C2	Critical	PAT encryption key falls back to public constant (ZAIMEM_SECRET unset)	Cryptography	Fixed
H1	High	API tokens stored in plaintext in the database	Credential storage	Fixed
H2	High	30 dependency advisories incl. Next.js middleware bypass (critical)	Supply chain	Open
H3	High	Unauthenticated token issuance with no rate limiting	Abuse resistance	Fixed
M1	Medium	Unbounded JSON-RPC batch processing enables CPU/memory exhaustion	Abuse resistance	Fixed
M2	Medium	Token accepted via URL query string (log and Referer leakage)	Credential handling	Documented
M3	Medium	MCP session registry grows without eviction	Resource safety	Fixed
M4	Medium	CORS wildcard origin with credential headers exposed	Transport	Documented
M5	Medium	Verbose internal error messages returned to clients	Info disclosure	Documented
L1	Low	No security headers (nosniff, frame options, referrer policy)	HTTP hardening	Fixed
L2	Low	TypeScript build errors ignored (ignoreBuildErrors)	Build hygiene	Open
L3	Low	Token persisted in localStorage (XSS impact amplifier)	Client security	Documented
L4	Low	9 moderate dependency advisories (uuid, js-yaml, flatted)	Supply chain	Open

Table 1: Consolidated findings register. "Fixed*" for C1 means removed from HEAD and pushed; the tokens themselves must still be revoked by their owner (Section 4.1).

Status legend: **Fixed** - remediation implemented and verified in commit 4c00866. **Documented** - risk assessed and consciously accepted for the current deployment model, with compensating context documented in the corresponding finding section. **Open** - remediation recommended but requiring either a dependency upgrade window or a maintainer decision.

4. Critical Findings

4.1 C1 - Live GitHub tokens committed to the public repository

Component: scripts/e2e-test.ts, scripts/e2e-github-unit.ts. **Risk: critical, credential compromise.** The end-to-end test suite contained two real GitHub personal access tokens as string literals. Git history shows they entered the repository with the v1.1 commit (f357cee) and remained present through v1.7.1 (0669d87), both of which are on origin/main of the public github.com/romangalaxys10-spec/zaimem repository. One of the tokens was functional against the GitHub API and was used by the rescue-import test to exercise a live cloud-database repository; the second was an invalidated token used to drive the negative pairing path.

Anyone browsing the public repository - or an automated secret-scanning crawler - could read these tokens directly from the source. A functional PAT grants whatever scopes it was created with, potentially including read access to every private repository of the owning account. This is the classic hard-coded-credential failure mode (CWE-798), and the exposure window measured in months.

Remediation. All nine occurrences were replaced with a deterministic local-only dummy token; the previously fake-mocked rescue-import scenario is now env-gated: when ZAIMEM_E2E_PAT (and optionally ZAIMEM_E2E_REPO) are exported, the live import path is exercised; without them the nine affected checks are emitted as hermetic skips, keeping the suite green without any secret in source. Verified: the sanitisation script reports zero real-looking tokens remaining, and the full e2e suite passes 113/113. **The tokens themselves must still be revoked on GitHub** - deleting them from HEAD does not un-leak them from history, and history rewriting (BFG Repo-Cleaner or git filter-repo) is only advisable after revocation.

4.2 C2 - PAT encryption key falls back to a public constant

Component: src/lib/zaimem/crypto.ts. **Risk: critical, cryptography (CWE-321).** ZaiMem stores paired GitHub PATs AES-256-GCM encrypted, with the key derived via script from the ZAIMEM_SECRET environment variable. The implementation, however, fell back to the hard-coded string "zaimem-default-secret-rotate-me" when the variable is unset - and inspection of the runtime environment showed it was indeed unset. The sandbox .env contained only DATABASE_URL, so every stored PAT was encrypted under a key that is public in the source repository. The cipher was correct, but the key material made the exercise equivalent to obfuscation: anyone with the SQLite file and the public source could decrypt all user PATs.

Remediation. A cryptographically random 256-bit ZAIMEM_SECRET was generated and installed in the environment. Because existing rows were encrypted under the legacy key, the decryption path now implements a transparent upgrade: decryptSecretUpgradable tries the current key first and falls back to the

legacy key only for pre-1.7.2 payloads, and the shared loadPat accessor in github.ts re-encrypts such payloads under the current key on first use. A startup warning now fires if the secret is missing, so a misconfigured deployment is visible in logs. **Deployment note:** the production instance at zaimem.space-z.ai must also receive ZAIMEM_SECRET in its environment when build 4c00866 is deployed; until then it will run under the legacy key with the warning visible.

5. High-Severity Findings

5.1 H1 - API tokens stored in plaintext

Component: prisma/schema.prisma, src/lib/zaimem/auth.ts. **Risk: high, credential storage.** The ZaiMem token (zm_ + 48 hex chars, 192 bits of entropy) is the complete account credential: it authenticates every MCP and REST call. The database stored it verbatim, and every lookup scanned the plaintext column. A database leak - a copied db/custom.db, a misconfigured backup, an exposed snapshot in the cloud-database repo - would have exposed every active token in usable form, converting a data breach into immediate account takeover.

Remediation. Tokens are now stored only as SHA-256 hashes in a new unique tokenHash column; the plaintext legacy column is nullable and is blanked on migration. Lookup happens exclusively by hash, with a lazy migration path that upgrades any legacy row on its first authenticated use; a one-off backfill script additionally migrated all 53 existing rows immediately, verified by a database query showing zero plaintext tokens remaining. Session prompts that previously read the stored token to embed into bootstrap prompts were rewired to use the request-scoped token, so no code path needs the plaintext at rest. The issuance response remains the only moment the plaintext token ever exists server-side.

5.2 H2 - Dependency vulnerabilities including a critical Next.js advisory

Component: package.json / lockfile. **Risk: high, supply chain.** The advisory scan reported 30 vulnerabilities: 2 critical, 17 high, 9 moderate and 2 low. The criticals are a Next.js middleware/proxy bypass via segment-prefetch routes in App Router applications and an Auth.js (next-auth) email-normaliser homoglyph bypass. High-severity items include sharp (inherited libvips and libheif CVEs - reachable if user-supplied images are processed), lodash and lodash-es template code injection, defu and flatted prototype pollution, js-yaml quadratic-complexity denial of service, nanoid integer overflow and postcss arbitrary file read via source mappings.

The Next.js middleware bypass is partially mitigated by the fact that ZaiMem defines no middleware file - authorisation is enforced inside every route handler - so the classic bypass target is absent. sharp is an optional transitive dependency and the image pipeline does not invoke it on the audited path. Nevertheless, the critical and high set warrants an upgrade window: update Next to the patched release, bump next-auth or confirm it is unused at runtime, replace or update sharp, and run npm audit fix for the non-breaking set. These are tracked as open remediation items R2 and R3 in Section 9.

5.3 H3 - Unauthenticated account issuance without rate limiting

Component: `src/app/api/auth/init/route.ts`. **Risk: high, abuse resistance.** The passwordless model issues a full account to any anonymous POST. Before this audit the endpoint had no rate limiting of any kind: a script could create unbounded users, seeding builtin skills per creation and growing the database without limit - a trivial resource-exhaustion and storage-abuse vector on a publicly reachable endpoint. The same absence extended across the API surface: no endpoint applied any form of throttling.

Remediation. A fixed-window in-memory rate limiter (`ratelimit.ts`) was introduced and wired at the two highest-risk boundaries: `/api/auth/init` is capped at 60 issuances per 5 minutes per client IP (proxy-aware via `x-forwarded-for`), and the MCP endpoint is capped at 1,200 calls per minute per token - generous enough that no legitimate agent workflow approaches it, tight enough to stop sustained scripted abuse. Limits deliberately sit above the e2e suite consumption profile, verified by the 113/113 green run. For multi-instance deployments the limiter should be swapped for a shared store such as Redis; the interface isolates this change to one module.

6. Medium-Severity Findings

6.1 M1 - Unbounded JSON-RPC batch processing

Component: `src/lib/zaimem/mcp.ts`, `handleMcpPost`. **Risk: medium, denial of service.** The MCP transport accepts JSON-RPC arrays and dispatches every element serially. Without a batch cap, a single request containing tens of thousands of trivial notifications forced the server to execute them all within one HTTP request, consuming CPU and memory while blocking the event loop window. Batch semantics amplify every other per-call cost, including per-tool pack preference lookups.

Remediation. Batches are now capped at 25 requests; larger arrays are rejected with a JSON-RPC `-32600 Invalid Request` error before any dispatch. The limit matches the MCP spec guidance for batch sizing and comfortably covers every observed client pattern. Combined with the per-token rate limit (H3), the worst-case per-token work per minute is now bounded.

6.2 M2 - Token accepted via URL query string

Component: `src/lib/zaimem/auth.ts`, `extractToken`. **Risk: medium, credential leakage.** For client convenience the API accepts the token as a query parameter (`?token=zm_...`) on every endpoint. Query strings leak into access logs, proxy logs, browser history and the Referer header of any outbound link clicked from a token-bearing page. The magic prompt generated for agents additionally documents the query-string form, so real deployments will contain tokens in URLs.

Assessment: documented, accepted for now. The header form is the primary and recommended authentication path; the query form exists because some MCP clients cannot set custom headers. Tokens are random 192-bit values, rotation is cheap, and the deployment does not emit access logs to shared storage. Recommendation R6 proposes deprecating the query variant in client docs and adding a server toggle to

reject it once client support allows.

6.3 M3 - MCP session registry never evicted

Component: `src/lib/zaimem/mcp.ts`, `mcpSessions` map. **Risk: medium, resource exhaustion.** Every JSON-RPC initialize call inserts an entry into an in-memory session map that had no eviction path and no size cap. Over weeks of uptime with automated agents re-initialising, the map grows monotonically - a slow memory leak reachable by unauthenticated... rather, by any token holder issuing repeated initialises, and amplified by the prior absence of rate limiting.

Remediation. Registry entries now expire after 24 hours, enforced at insertion time: when the map exceeds 10,000 entries, stale entries are pruned and the oldest entries are evicted until under the cap. Session identity remains advisory (authentication is token-based per request), so eviction has no authorisation impact - verified by the full e2e suite, which re-initialises sessions extensively.

6.4 M4 - Permissive CORS with credential headers

Component: `src/lib/zaimem/auth.ts`, `CORS_HEADERS`. **Risk: medium, transport policy.** The MCP endpoint responds with `Access-Control-Allow-Origin: *` and explicitly allows `Authorization` and `X-Zaimem-Token` headers cross-origin. Any web page can therefore issue credentialed-looking requests to the API from a victim browser; only the token value itself protects the account. Because the application uses no cookies and the token must be explicitly attached, the wildcard does not by itself enable cross-site requests with ambient credentials - but it widens the blast radius of any separately leaked token (for example via M2) by allowing browser-based abuse from any origin.

Assessment: documented, accepted. Open CORS is a deliberate design choice for an MCP server that third-party agent runtimes call from arbitrary origins; restricting origins would break the core product promise. The compensating controls are the token entropy, the new rate limits, and the batch cap. If a fixed set of partner origins emerges, `Allow-Origin` should switch to an allow-list (R7).

6.5 M5 - Verbatim internal error messages

Component: MCP fatal handler, `/api/github fail()`. **Risk: medium, information disclosure.** Uncaught exceptions are returned to clients with `err.message` embedded in the JSON-RPC error data, and the GitHub route forwards `GhError` text plus raw exception strings. Internal messages can reveal stack-adjacent detail - file paths, upstream API responses, dependency internals - that assists an attacker in mapping the backend.

Assessment: documented. Reviewed call sites only surface `GhError` messages (already user-directed) or generic exception text; no secrets, tokens or database contents are included, and the development server stacks are not part of production responses. The finding is retained as a hardening item: centralise error serialisation so production responses use stable, low-detail messages while keeping detail in server logs (R8).

7. Low-Severity and Informational Findings

7.1 Low-severity findings

L1 - Missing security headers (fixed). The application set no X-Content-Type-Options, X-Frame-Options, Referrer-Policy, Permissions-Policy or COOP. `next.config.ts` now applies all five globally. A strict Content-Security-Policy is deliberately deferred (R5) because the dashboard uses inline styles extensively; a nonce-based CSP is the correct follow-up.

L2 - TypeScript build errors ignored (open). `typescript.ignoreBuildErrors = true` in `next.config.ts` trades type safety for build resilience. Type errors that would be caught at build time can instead surface as runtime faults in production, occasionally in security-relevant paths. Recommend clearing the backlog and re-enabling strict builds.

L3 - Token in localStorage (documented). The dashboard stores the account token in `localStorage`. Any successful XSS becomes full account takeover. With no third-party scripts, escaped React rendering and a tightened header set the practical risk is low; the token-as-account key model makes some client persistence unavoidable. A CSP plus a session-scoped storage option are the medium-term mitigations.

L4 - Moderate dependency advisories (open). Nine moderate advisories (uuid buffer bounds, js-cookie attribute injection, minimatch and picomatch ReDoS, brace-expansion and browserslist memory growth, others) are bundled in the same upgrade window as H2; none is remotely exploitable on the audited code path.

7.2 Verified positive controls

The audit also confirmed a meaningful set of controls that materially reduce risk and deserve explicit acknowledgment:

- **No SSRF on web tools.** `zaimem_web_fetch` executes through the hosted SDK `page_reader` service, not a raw server-side fetch; internal-network URLs never reach a ZaiMem-controlled HTTP client.
- **Expression evaluation is parser-based.** `zaimem_calc` tokenises input with a strict numeric grammar (no `eval`, no Function constructor); arbitrary characters are rejected before parsing.
- **Repository import validation.** `importFromRepo` enforces an owner/name regex after normalisation, blocking path traversal and URL injection into GitHub API paths.
- **Upload limits.** Document ingestion caps uploads at 10 MB and text payloads at a fixed character budget, with extension-aware extraction and rejection of unsupported types.
- **Uniform authorisation.** All 13 REST route groups and the MCP dispatcher authenticate through one helper and scope every Prisma query by `userId`; no cross-tenant object access was found (no IDOR).
- **Server-side tool-pack enforcement.** Disabled packs are enforced at `tools/call` in addition to `tools/list`, so clients cannot bypass dashboard toggles; locked packs cannot be disabled at all.

- **Repository commandeering guard.** ensurePrivateRepo refuses to reuse non-private repos or repos whose description does not identify ZaiMem management.
- **Hygiene.** .gitignore correctly excludes .env, logs and the database; no secrets or credentials are tracked other than the C1 tokens; token generation uses 24 random bytes (192 bits).

8. Remediation Applied and Verification

All code-level fixes were implemented during the engagement and pushed to the public repository as commit **4c00866 (v1.7.2)**. The table below maps each fix to its primary file.

Fix	Change	Primary files
C1	Replaced 9 real-token literals with a local-only dummy; rescue-import test env-gated via ZAIMEM_E2E_PAT / ZAIMEM_E2E_REPO with 9 hermetic skips	scripts/e2e-test.ts, scripts/e2e-github-unit.ts
C2	ZAIMEM_SECRET generated and installed; legacy-key decrypt with transparent re-encryption (auto-heal); missing-secret startup warning	src/lib/zaimem/crypto.ts, src/lib/zaimem/github.ts
H1	SHA-256 token hashing at rest (tokenHash unique); lazy migration plus one-off backfill of 53 rows; prompts rewired to request token	prisma/schema.prisma, src/lib/zaimem/auth.ts, src/app/api/auth/*, src/lib/zaimem/mcp.ts
H3	Fixed-window rate limiter: init 60/5min per IP; MCP 1,200/min per token	src/lib/zaimem/ratelimit.ts, src/app/api/auth/init/route.ts, src/lib/zaimem/mcp.ts
M1	JSON-RPC batch capped at 25 with -32600 rejection	src/lib/zaimem/mcp.ts
M3	Session registry eviction: 24h expiry, 10k entry cap	src/lib/zaimem/mcp.ts
L1	Global security headers: nosniff, SAMEORIGIN, referrer-policy, permissions-policy, COOP	next.config.ts
H1*	Export route token masking read the stored column; rewired to request token to survive hashing	src/app/api/memories/export/route.ts

Table 2: Fixes delivered in commit 4c00866.

8.1 Verification evidence

The complete end-to-end suite was executed against the hardened build after a Prisma schema push, client regeneration and server restart: **113 of 113 checks passed**, including the MCP protocol negotiation, all 33 tools, tool-pack gating, headroom toggling and the env-gated rescue-import skips. Database inspection confirmed **0 plaintext tokens remaining** after the backfill and successful authentication round-trips through both the hashed-lookup and legacy migration paths. A startup log review confirmed the new ZAIMEM_SECRET is loaded (no legacy-key warning). Rate limits and the batch cap were smoke-tested directly: issuances under the threshold return 200, and oversized batches are rejected with -32600 before dispatch.

9. Residual Risk and Recommendations

The following actions remain open, ordered by priority. R1 is urgent and owner-only; the rest schedule into the normal development cycle.

R1 - Revoke the leaked GitHub tokens (urgent, owner action).

In GitHub Settings, delete both PATs that were committed between v1.1 and v1.7.1 and review the account security log for their use. GitHub secret scanning may have already auto-revoked the classic token; verify rather than assume. After revocation, optionally purge the blobs from git history with git filter-repo or BFG.

R2 - Deploy the hardened build with its secret.

Ship 4c00866 to production with ZAIMEM_SECRET set in the production environment. Users with paired GitHub links re-encrypt automatically on next use; a re-pair is only needed if the legacy fallback is removed later.

R3 - Dependency upgrade window.

Update Next.js to the patched release for the middleware-bypass advisory, align next-auth, update or drop sharp, and apply npm audit fix for the non-breaking set (H2, L4). Re-run the e2e suite after the upgrade.

R4 - Re-enable strict TypeScript builds.

Clear the ignoreBuildErrors backlog and enforce the build gate in CI (L2).

R5 - Content-Security-Policy.

Introduce a nonce-based CSP once dashboard inline styling is refactored; this directly reduces the L3 localStorage exposure window.

R6 - Deprecate query-string tokens.

Announce header-only authentication for MCP clients and add a server-side toggle to reject ?token= once client coverage allows (M2).

R7 - Origin policy review.

If partner origins stabilise, replace the CORS wildcard with an allow-list while keeping the MCP public contract (M4).

R8 - Central error serialisation.

Return stable low-detail messages in production and keep diagnostics in server-side logs (M5).

R9 - Shared rate-limit store.

Move the in-memory limiter behind Redis or equivalent when scaling beyond one instance; the module interface already isolates this swap (H3).

With R1 completed and the remaining items scheduled, ZaiMem's security posture is solid for its deployment model: the credential vault is now genuinely encrypted, credentials at rest are hashed, the public attack surface is rate-limited and bounded, and the test infrastructure no longer carries live secrets. The audit confirms that the per-user authorisation model - the core of a multi-tenant memory service - held up under review with no cross-tenant access path found.

9.1 Audit artefacts

The engagement produced reusable artefacts beyond this report: a token-sanitisation script with a no-secrets regression guard, a one-off hash backfill script (retained as documentation of the migration), the rate-limiter module now part of the application, and the two chart assets used in Section 3. These artefacts are committed with the v1.7.2 hardening so the next audit cycle can re-run the same checks deterministically.

Artefact	Purpose	Reference
Hardening commit	All v1.7.2 security fixes pushed to origin/main	4c00866
E2E verification	Full regression suite against the hardened build	113 / 113 checks passed
Token backfill	Legacy plaintext tokens migrated to SHA-256 hashes	53 rows, 0 remaining
Sanitisation guard	CI-ready check that no real PAT re-enters source	scripts/sec-fix-sanitize-pats.py
Rate limiter	Reusable fixed-window limiter (auth + MCP wired)	src/lib/zaimem/ratelimit.ts

Table 3: Audit artefacts and verification references.